



Book I. Prop. XXVIII. Theor. *If a straight line (—), cutting two other straight lines (— and —), makes the external equal to the internal and opposite angle, at the same side of the cutting line (namely $\blacktriangle = \triangle$ or $\triangle = \triangle$), or if it makes the two internal angles at the same side ($\triangle$ and $\triangle$, or $\triangle$ and $\triangle$) together equal to two right angles, those two straight lines are parallel.*

First, if $\blacktriangle = \triangle$, then

$\blacktriangle = \triangle$ (pr. I.15),

$\therefore \triangle = \triangle$

$\therefore \text{—} \parallel \text{—}$ (pr. I.27).

Secondly, if $\triangle + \triangle = \text{semicircle}$,

then $\triangle + \triangle = \text{semicircle}$ (pr. I.13),

$\therefore \triangle + \triangle = \triangle + \triangle$ (ax. III)

$\therefore \triangle = \triangle$

$\therefore \text{—} \parallel \text{—}$ (pr. I.27).

Q. E. D.